The documentation and process conversion measures necessary to comply with this document shall be completed by 8 March 2022.

INCH-POUND

MIL-PRF-19500/643E
8 December 2021
SUPERSEDING
MIL-PRF-19500/643D
27 May 2020

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, UNITIZED, DIODE, SILICON, POWER RECTIFIER, ULTRAFAST, DUAL (COMMON CATHODE OR COMMON ANODE CENTER TAP) TYPES 1N6766 AND 1N6767, STANDARD AND REVERSE POLARITY, QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, high voltage, ultrafast, power rectifier diodes in a dual die, center-tap configuration. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package is a three terminal flange mount header configuration (TO-254AA, isolated) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25 °C (for each leg).

Types	V_{RWM} (1) I_D = 10 µA dc	I_{FSM} (1) t_P = 8.3 ms	I_F (1) (2) T_C = 100 °C	$R_{\theta JC}$ (1)	$R_{\theta JA}$ (1)	T_{STG} and T_J
	V dc	A (pk)	A dc	°C/W	°C/W	°C
1N6766, 1N6766R	400	125	12	1.8	40	-65 to +150
1N6767, 1N6767R	600	125	12	1.8	40	-65 to +150

(1) Each individual diode.
(2) Derate at 240 mA/°C above T_C = +100 °C.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics are at +25 °C, and for each diode.

Types	V_{F1} I_F = 6 A dc	V_{F2} I_F =12 A dc	I_{R1} (see 1.3) V_R = 0.8 V_{RWM}	I_{R2} V_R = 0.8 V_{RWM} T_C = +100 °C	t_{rr}	C_J V_R = 5 V f = 1 MHz
	V dc	V dc	µA dc	µA dc	ns	pF
1N6766, 1N6766R	1.35	1.55	10	1,000	60	300
1N6767, 1N6767R	1.35	1.55	10	1,000	60	300

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN brand and quality level designators for encapsulated devices. The quality level designators for devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Device type. The designation system for the device types of diodes covered by this specification sheet are as follows.

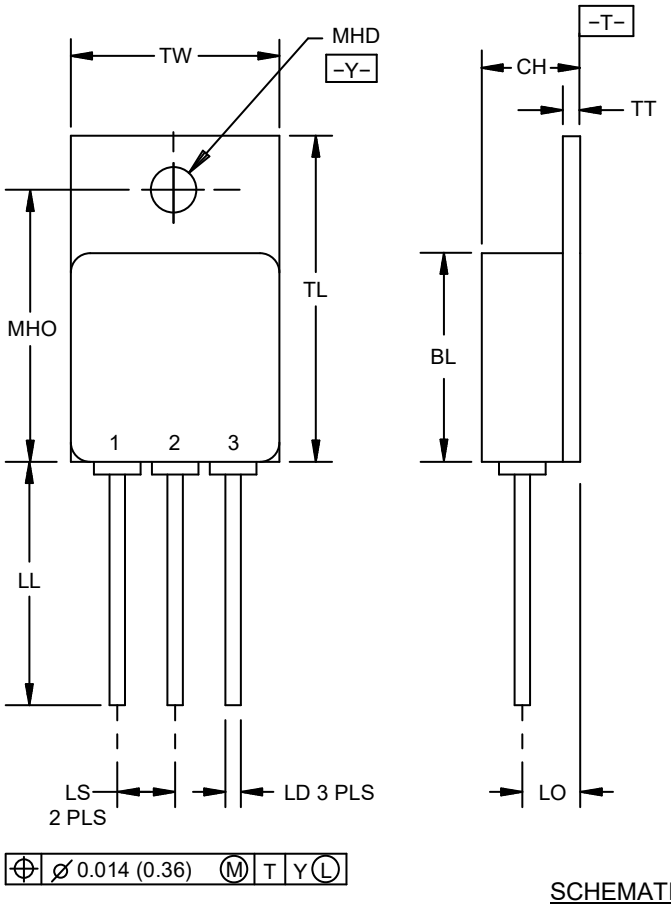
1.5.2.1 First number and first letter symbols. The diodes of this specification sheet are identified by the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the diodes covered by this specification sheet are "6766 and "6767".

1.5.3 Suffix letter. The suffix letter "R" is used on devices that are reverse polarity (see figure 1). No suffix letter is used on devices that are of standard polarity.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

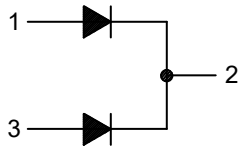
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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84

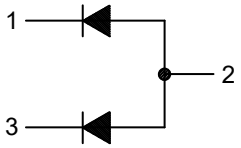
SCHEMATICS

1N6766, 1N6767



TERM 1 = ANODE 1
TERM 2 = CATHODE
TERM 3 = ANODE 2

1N6766R, 1N6767R



TERM 1 = CATHODE 1
TERM 2 = ANODE
TERM 3 = CATHODE 2

NOTES:
1. Dimensions are in inches. Millimeters given for general information only.
2. Protrusion of ceramic eyelets included in dimension LL.
3. All terminals are isolated from case.
* 4. In accordance with ASME Y14.5, diameters are equivalent to Øx symbology.

FIGURE 1. Dimensions and configuration of three terminal flange mount header package (TO-254AA, isolated).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Lead formation. Where a choice of lead formation is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of table E-IV of MIL-PRF-19500 and 100 percent dc testing in accordance with subgroup 2 of table I herein on all devices that had its leads altered.

3.4.3 Lead isolation. Methods used for electrical isolation of the terminal feed through shall employ materials that contain a minimum of 90 percent ceramic AL_2O_3 or equivalent. Examples of such construction techniques are metalized ceramic eyelets or ceramic walled packages.

3.4.4 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (quality levels JANTX, JANTXV, and JANS only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.1)	Thermal impedance (see 4.3.1)
9 and 10	Not applicable	Not applicable
11	I_{R1} and V_{F1}	I_{R1} and V_{F1}
12	See 4.3.2, t = 240 hours	See 4.3.2, t = 48 hours
13	Subgroups 2 and 3 of table I herein; V_{F1} and I_{R1}; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 µA, whichever is greater; $V_{F1} \leq \pm$100 mV.	Subgroup 2 of table I herein; V_{F1} and I_{R1}; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 µA whichever is greater; $\Delta V_{F1} \leq \pm$100 mV.
17	Case isolation. Method 1081 of MIL-STD-750	Case isolation. Method 1081 of MIL-STD-750

(1) Thermal impedance shall be performed any time after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.

4.3.1 Thermal impedance The thermal impedance measurements shall be performed in accordance with method 3101 or 4081, as applicable, of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (V_C and V_H where appropriate). See table II, subgroup 4 herein.

4.3.2 Power burn-in conditions. Power burn-in shall be performed in accordance with test condition A of method 1038 of MIL-STD-750. The following details shall apply: T_C = +125 °C; V_R = 0.8 of rated V_{RWM} (see 1.3).

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX and JANTXV. Delta measurements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

Subgroup	Method	Condition
B4	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85 °C minimum, for 2,000 cycles minimum.

4.4.2.2 Quality levels JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500).

Subgroup	Method	Condition
B3	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85 °C minimum, for 2,000 cycles minimum.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Delta measurements shall be in accordance with the applicable steps of table III herein.

Subgroup	Method	Condition
C2	2036	Test condition A, 5 pounds (2.27 kg), t = 15 seconds ±3 seconds.
C5	4081	$R_{\theta JC}$ (maximum) = 1.8 °C/W.
C6	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85 °C minimum, for 6,000 cycles minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and in table II herein. Delta measurements shall be in accordance with the applicable steps and footnotes of table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 µA/division and 50 to 100 V/division. Reverse current over the knee shall be at least 500 µA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

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* TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.1	$Z_{\theta JX}$			°C/W
* Breakdown voltage 3/	4021	I_R = 10 µA dc, pulsed 3/	V_{BR}			V dc
* 1N6766, 1N6766R				440		
* 1N6767, 1N6767R				660		
* Forward voltage 3/	4011	Condition B, pulsed 3/				
		I_F = 6 A dc	V_{F1}		1.35	V dc
		I_F = 12 A dc	V_{F2}		1.55	V dc
* Reverse leakage current 3/	4016	Condition A or B; pulsed 3/ V_R = 0.8 of V_{RWM} (see 1.3)	I_{R1}		10	µA dc
Subgroup 3						
High temperature operation:		T_C = +100 °C				
Reverse leakage current 3/	4016	Condition A or B; pulsed V_R = 0.8 of V_{RWM} (see 1.3)	I_{R2}		1.0	mA dc
Low temperature operation:		T_A = −55 °C				
Forward voltage 3/	4011	Condition B, I_F = 12 A dc, pulsed	V_{F3}		1.70	V dc
Subgroup 4						
Scope display evaluation		See 4.5.2				
Reverse recovery time measurements	4031	Condition B1	t_{rr}		60	ns
Subgroups 5 and 6						
Not applicable						

See footnotes at end of table.

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* TABLE I. Group A inspection − Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Junction capacitance	4001	V_R = 5 V dc; f = 1.0 MHz	C_J		300	pF

1/ For sampling plan, see MIL-PRF-19500.
2/ Each individual diode.
3/ Pulse test: Pulse width = 300 µs, duty cycle ≤ 2 percent.

TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices, c = 0
Temperature cycling	1051	500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, and 6.	
Subgroup 2			22 devices, c = 0
Steady-state reverse bias	1038	Test condition A, t = 1,000 hours, T_C = +125 °C V_R = 0.8 of rated V_{RWM} (see 1.3).	
Electrical measurements		See table III herein, steps 1, 2, and 6.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See subgroup 4, table E-IX of MIL-PRF-19500.	
Subgroup 5			15 devices, c = 0
Barometric pressure, (reduced)	1001	Condition D, pressure 8.0 mm Hg.	

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TABLE III. Electrical and delta measurements for groups A, B, C, and E. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Forward voltage	4011	Condition B, I_F = 12 A dc, pulsed	V_{F2}		1.55	V dc
2	Reverse leakage current	4016	Condition A or B, pulsed, V_R = 0.8 of percent V_{RWM} (see 1.3)	I_{R1}		10	µA dc
3	Forward voltage	4011	Condition B, pulsed, I_F = 12 A dc	ΔV_{F2}	±100 mV dc from initial reading.		
4	Reverse leakage current	4016	Condition A or B, pulsed V_R = 0.8 of percent V_{RWM} (see 1.3)	ΔI_{R1}	100 percent of initial value or ±2.5 µA dc whichever is greater.		
5	Thermal impedance	3101	See 4.3.1	$Z_{\theta JX}$		1.60	°C/W
6 5/	Reverse recovery time	4031	See table I, subgroup 4 herein	t_{rr}		60	ns

1/ The electrical measurements taken during group B inspection for quality level JANS (table E-VIA of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 4, the measurements of steps 1, 2, 3, 4, 5, and 6 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 5, the measurements of steps 1, 2, 3, 4, and 6 of this table shall also be taken.

2/ The electrical measurements taken during group B inspection for quality level JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1, 2, 5, and 6 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of steps 1, 2, 3, 4, 5, and 6 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 of this table shall also be taken.

3/ The electrical measurements taken during group C inspection for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1, 2, and 5 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 6, the measurements of steps 1, 2, 3, 4, 5, and 6 of this table shall also be taken.

4/ The electrical measurements taken during group E inspection for all quality levels (table E-IX of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 1, the measurements of steps 1, 2, 3, 4, 5, and 6 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 of this table shall also be taken.

5/ Step 6 applies to irradiated devices only.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

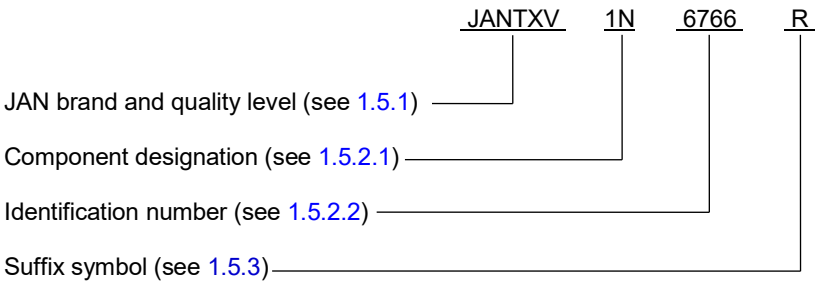
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1) and if applicable, any needed lead formation (see 3.4.2).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are construction using the following form.



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6.5 List of PINs. The following is a list of possible PINs for the devices available on this specification sheet.

Base quality level	Quality level "TX"	Quality level "TXV"	Quality level "S"
Standard polarity			
JAN1N6766	JANTX1N6766	JANTXV1N6766	JANS1N6766
JAN1N6767	JANTX1N6767	JANTXV1N6767	JANS1N6767
Reverse polarity			
JAN1N6766R	JANTX1N6766R	JANTXV1N6766R	JANS1N6766R
JAN1N6767R	JANTX1N6767R	JANTXV1N6767R	JANS1N6767R

6.6 Cross reference list. Parts covered by this specification may be used to replace the following commercial PIN:

Preferred types	Commercial types
JAN1N6766, JAN1N6766R JANTX1N6766, JANTX1N6766R JANTXV1N6766, JANTXV1N6766R JANS1N6766, JANS1N6766R	1N6766, 1N6766R
JAN1N6767, JAN1N6767R JANTX1N6767, JANTX1N6767R JANTXV1N6767, JANTXV1N6767R JANS1N6767, JANS1N6767R	1N6767, 1N6767R

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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